

Ian Marcus Breganza

Machine Learning Engineer · Full-Stack Developer

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SUMMARY

CS graduate with hands-on experience building AI systems and web applications. Trained an image classification model reaching 99% accuracy; built and deployed a face recognition attendance system on a small single-board computer; and developed secure web features during a government IT internship. Comfortable working across both AI and web development, from building models to backend systems.

EDUCATION

Bachelor of Science in Computer Science

National University, Laguna Campus

July 2026

EXPERIENCE

Freelance ML Engineer

Los Baños, Laguna

March 2026 – May 2026

Real-time Face Recognition Attendance System

- Built and deployed a face recognition attendance system on a Raspberry Pi 4 that automatically logs attendance by identifying people through a camera in real time.
- Developed the full workflow from camera input to attendance logging and delivered a working system to a client.
- Diagnosed a deployment latency bottleneck; optimized inference frame rates on Raspberry Pi 4 by leveraging lightweight models (YOLOv5n-Face) and documenting performance trade-offs for the next version of the system.

Student Researcher, Thesis Lead

National University Laguna | Calamba, Laguna

Apr 2024 – May 2026

Classification of Dipterocarp Species via Ensemble CNN Architecture

- Designed a dual-branch VGG16-based CNN fusing leaf and bark image modalities, achieving 99% classification accuracy on held-out test data across multiple Dipterocarp species.
- Applied transfer learning, data augmentation, and ensemble strategies on a small, field-collected dataset; evaluated robustness using precision, recall, and F1-score.
- Led a four-member research team coordinating dataset curation, code versioning using Git, and preparation of two submitted manuscripts: an IMRAD paper and an IEEE-formatted paper.

IT & Systems Development Intern

DOST – FPRDI | UPLB, College, Laguna

Dec 2025 – Feb 2026

PROMIS: Project Monitoring and Information System (Laravel / LAMP Stack)

- Built a digital certificate management module that lets admins upload and validate security certificates, with a dashboard showing expiration dates and upload history.
- Added password security features and automatic digital signature insertion into official reports; presented the completed system to department heads and the Executive Committee.
- Designed a training workshop for ~50 senior stakeholders (department heads, executives, scientists) covering system onboarding and official documentation.

TECHNICAL SKILLS

ML / AI: PyTorch, TensorFlow, Keras, scikit-learn, spaCy, RoBERTa, NumPy, Pandas, YOLOv5n-Face, ArcFace, OpenCV

Languages: Python, PHP, Java, JavaScript, HTML/CSS

Web / Backend: Laravel (LAMP Stack), FastAPI, REST APIs, AJAX, OpenSSL / PNPKI

Tools / Hardware: Git, VS Code, Kaggle, MS Office, Raspberry Pi 4, Docker